

| Question |     |      | Marking details |                                                                                                                                                                                                                                                                  | Marks available |           |          |           |           |           |
|----------|-----|------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|----------|-----------|-----------|-----------|
|          |     |      |                 |                                                                                                                                                                                                                                                                  | AO1             | AO2       | AO3      | Total     | Maths     | Prac      |
|          |     | (iv) | I.              | Gradient calculated correctly e.g. $\frac{12.4 - 0.6}{600} = 0.02$ [1]<br>$W_B = \text{gradient ecf} \times 980$ e.g. = approx 19.6 [N] [1]                                                                                                                      |                 |           | 2        | 2         | 2         | 2         |
|          |     |      | II.             | y-intercept determined correctly $\pm$ small square tolerance e.g. 0.7 [N] [1]<br>$W_R = 2 \times y\text{-intercept}$ e.g. = 1.4 N <b>ecf</b> but don't apply for intercept of 0 [1]                                                                             |                 |           | 2        | 2         | 2         | 2         |
|          | (d) |      |                 | % Unc (d) calculated i.e. $1 \times \frac{100}{100} = 1\%$ [1]<br>Uncertainty in (spread) $\Delta F = \frac{2.8 - 2.6}{2} = 0.1$ [1]<br>$p_F = \frac{0.1 \text{ecf}}{2.7} \times 100\% = 3.7\%$ [1]<br>Ignore s.f.                                               | 1               | 1         |          | 3         | 3         | 3         |
|          | (e) |      |                 | Any 1 $\times$ (1) of<br>[Spirit] level / appropriate method to ensure that the ruler is horizontal<br>Use digital forcemeter / forcemeter with a higher resolution<br>Clamp stand to hold newtonmeter<br>Repeat readings <u>of force</u> [due to random errors] |                 |           | 1        | 1         |           | 1         |
|          |     |      |                 | <b>Question 5 total</b>                                                                                                                                                                                                                                          | <b>3</b>        | <b>11</b> | <b>5</b> | <b>19</b> | <b>14</b> | <b>17</b> |